

MODEL CARD

GLM-4.6

An open-weight mixture-of-experts language model from Zhipu AI, documented from public sources — with every claim marked by how well it is actually sourced.

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At a glance

Developer	Zhipu AI, operating publicly as Z.ai
Released	Late September 2025
Architecture	Mixture-of-Experts Transformer, 355 B total parameters
Context window	200K tokens in, up to 128K tokens out
License	MIT — weights downloadable and commercially usable
Built for	Agentic task execution, software coding, long-context reasoning
Predecessor	GLM-4.5

How to read this card

Zhipu AI published a good deal about GLM-4.6 and stayed quiet about a good deal more. Rather than smooth over the difference, we tagged every substantive claim:

- [CONFIRMED] Stated directly in the developer’s own release material.
- [INFERRED] Reasoned from the documented GLM family pattern. Not stated outright for GLM-4.6, and not safe to cite as a specification.
- [NOT DISCLOSED] Never published. We left the field empty rather than borrowing a number from a comparable model.

The tags are the honest part of this card. A fair share of what a complete model card should contain is simply not in the public record, and we thought that was worth showing rather than hiding.

1 Purpose

GLM-4.6 is a general-purpose large language model, part of Zhipu AI's General Language Model series. It generates and reasons over natural language text for a broad range of downstream applications, with particular emphasis on three areas: **agentic task execution** (tool use and multi-step planning), **software coding** (generation, editing, and terminal-based tasks), and **long-context reasoning**. [CONFIRMED]

It is positioned as an open-weight alternative to proprietary frontier models. Because the weights are downloadable under an MIT license, enterprises and researchers can self-host, fine-tune, and deploy it without depending on a third-party API. The stated goal is to narrow the gap between open and closed models on real-world coding and agentic benchmarks while staying fully inspectable. [CONFIRMED]

2 Dataset

Zhipu AI has not published a detailed breakdown of GLM-4.6's pretraining corpus. This is normal practice for frontier-scale models, and we treat it as a transparency limitation rather than an oversight — see Section 6. From the developer's technical reports and the GLM family's documented lineage, three things can be said with reasonable confidence:

- The pretraining corpus is large-scale and multilingual, primarily Chinese and English web text, supplemented with code repositories, technical and academic documents, and dialogue data — consistent with GLM-4 and GLM-4.5. [INFERRED]
- GLM-4.6 builds on the GLM-4.5 architecture and training pipeline, with additional training extending context handling to 200K tokens and refreshing coding- and agent-relevant data: tool-use trajectories, terminal sessions, and front-end UI generation examples. [INFERRED]
- Post-training data includes curated instruction-following, coding, and agentic-workflow datasets for supervised fine-tuning, plus preference and reward data for reinforcement learning. [CONFIRMED]

Four fields that a complete dataset section would carry are absent from the public record entirely: **total corpus size**, **source distribution and weighting**, **filtering and deduplication method**, and **data licensing status**. [NOT DISCLOSED]

We looked for these and could not find them. If you come across a specific figure for GLM-4.6's dataset size in a blog post or comparison table, it is an estimate someone made — not a number Zhipu AI released.

3 Training method

The base architecture is a Mixture-of-Experts Transformer with 355 B total parameters, of which only a subset of experts activates per token — which is how a model this large stays affordable to run. [CONFIRMED]

The training pipeline, as far as it can be reconstructed, runs in four stages:

- **Pretraining.** Multi-stage training on the corpus described in Section 2. [INFERRED]

- **Context extension.** A dedicated phase stretching the input window to 200K tokens. [INFERRED]
- **Supervised fine-tuning.** Curated instruction, coding, and agentic-task data, teaching the base model to follow instructions and use tools reliably. [CONFIRMED]
- **Reinforcement learning.** Tuned for coding accuracy, agentic task completion, and reasoning stability. Zhipu AI reports this improved token efficiency by roughly **15%** — about 15% fewer tokens to finish comparable tasks than GLM-4.5 took. [CONFIRMED]

Compute is a blank. Hardware, total training FLOPs, and training duration were never published, so we have omitted them rather than estimating from parameter count. [NOT DISCLOSED]

4 Evaluation

Zhipu AI reports GLM-4.6’s performance through its own benchmark suites rather than against a single certification standard. The main axes are:

- **CC-Bench**, an internally developed, human-evaluated real-world coding benchmark. GLM-4.6 is reported to reach near-parity with Claude Sonnet 4, and the developer’s own materials acknowledge it still trails Claude Sonnet 4.5 on coding. [CONFIRMED]
- **Token efficiency**, measured as tokens consumed per completed task — roughly 15% fewer than GLM-4.5 for comparable work. [CONFIRMED]
- **General capability** benchmarks covering reasoning, mathematics, and multilingual understanding, consistent with the broader GLM-4.x suite. Public GLM-4.6-specific numbers are thinner here than for the GLM-4.6V multimodal variant. [INFERRED]

Model	Availability	Win rate
Claude Sonnet 4.5	Proprietary, Anthropic	97%
Claude Sonnet 4	Proprietary, Anthropic	90%
GLM-4.6	Open weight, Zhipu AI	88%
GLM-4.5	Open weight, Zhipu AI	76%

These are the developer’s own numbers, scored by evaluators the developer engaged, and no independent lab has reproduced them. That does not make them wrong — a company conceding that a competitor beats its own model is not the behaviour of someone gaming a benchmark — but a comparative claim nobody outside the company has checked is still an unchecked claim.

5 Bias analysis

There is no quantitative bias or fairness audit for GLM-4.6. No demographic parity testing, no BBQ or StereoSet result, nothing equivalent — from the developer or from anyone else. [NOT DISCLOSED]

That absence is the finding. What follows is inference from how the model was built, and we want to be clear that these are *reasoned risks, not measurements*:

- **Language and cultural reach.** Training data drawn predominantly from Chinese- and English-language web sources will likely produce stronger, more culturally attuned output in those languages, and weaker, more error-prone output for lower-resource languages and the cultural contexts attached to them. [INFERRED]
- **Inherited web-scrape skew.** As with essentially every large web-trained model, GLM-4.6 likely carries the representational imbalances of its sources — some demographic groups, viewpoints, and regions over-represented, others thin on the ground — which can surface as skewed associations in generated text. [INFERRED]
- **What the tuning prioritised.** Post-training emphasised coding and tool use. Conversational and creative-writing bias behaviour has probably had less dedicated attention than it would in a model built primarily for chat. [INFERRED]

If you are putting this model in front of users, run your own bias evaluation on your own population first. That is ordinary good practice with any model; here it is the only evidence that will exist.

6 Ethical considerations

- **The transparency gap.** Training data composition, red-teaming methodology, and the reinforcement learning reward criteria are not disclosed. This is what makes independent verification of any safety or fairness claim impossible, and it is the thread running through Sections 2, 4, and 5. [NOT DISCLOSED]
- **Open weights cut both ways.** A permissive MIT license with downloadable weights means whatever safety tuning ships with the model can be fine-tuned back out by anyone downstream. Openness is genuinely valuable, and it relocates responsibility for safe deployment onto whoever is hosting. [CONFIRMED]
- **Agentic capability raises the stakes.** A model designed for tool use, terminal access, and multi-step execution can act on a mistake rather than merely state one. Production deployments warrant a human in the loop. [INFERRED]
- **Jurisdiction is not inherited.** As a model from a China-based developer, deployments in some regions may need separate evaluation of data governance, export-control, and content-moderation requirements. [INFERRED]
- **Say when it is AI.** Output used in academic, legal, or journalistic work should be disclosed as AI-generated or AI-assisted — including, for the avoidance of doubt, work like this. [CONFIRMED]

7 Limitations

- **The documentation itself.** Dataset composition, training compute, and safety evaluation methodology are not fully disclosed, which caps how complete Sections 2, 4, and 5 of this card can be. [NOT DISCLOSED]
- **It is not the best coding model available.** Zhipu AI's own disclosures place GLM-4.6 behind the strongest proprietary coding models, Claude Sonnet 4.5 among them. [CONFIRMED]
- **Open does not mean cheap.** At 355 B parameters, even with sparse activation, self-hosting needs serious GPU infrastructure. The license is free; the hardware is not. [CONFIRMED]

- **The usual LLM caveats.** Hallucination, sensitivity to how a prompt is phrased, uneven performance well outside the coding and agentic domain it was tuned for, and a fixed knowledge cutoff. [INFERRED]
- **Nobody independent has checked.** At the time of writing we found no major third-party audit or standardised red-teaming report for GLM-4.6 in public sources. [NOT DISCLOSED]

8 Appropriate use

Well suited for

- Software development assistance — code generation, debugging, refactoring, and terminal or agentic coding workflows.
- Front-end and UI generation, where the model has been specifically noted for polished output.
- Long-document analysis and summarisation, using the 200K-token context window.
- Self-hosted deployments needing data privacy, on-premise control, or fine-tuning on a proprietary codebase — the case the MIT license exists to serve.
- Research into open-weight MoE behaviour and agentic AI systems.

Needs additional safeguards

- High-stakes autonomous decisions in medical, legal, or financial settings. No domain-specific safety evaluation has been published, so human review is not optional.
- Deployments serving low-resource languages or underrepresented cultural contexts, without evaluation of your own.
- Fully autonomous action with real-world consequences — payments, system administration, irreversible file operations — without a human in the loop.
- Anything requiring an audited fairness guarantee. This one cannot be satisfied at all right now, because no such audit is public.

Closing note

GLM-4.6 is a genuinely open model, released under a license that means what it says, and benchmarked candidly enough by its developer to admit a competitor is ahead. Those are real virtues and they are not universal.

The gap is documentation. Data provenance, training compute, and fairness evaluation are the three areas where an open-weight release most needs a paper trail — because the license hands capability to people who will run the model in contexts its developer never sees — and those are precisely the three areas left blank. The weights are open; the record behind them is not.

Prepared as an academic exercise from publicly available secondary sources — developer blog posts, technical press coverage, and published benchmark reports — as of August 2026. Not an official Zhipu AI publication. Where we could not find a primary source, we said so instead of guessing.